Credit Portfolio Risk Analysis

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Portfolios: 1 & 2

Course: Introduction to Credit Risk and Applications in Python

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1. Introduction

This report is being done under the scope of the course Introduction to Credit Risk and Applications in Python and presents a comprehensive analysis of the one year credit risk of two given portfolios using a controlled simulation.

The two models used for this analysis are: **Model A (The One-Factor Systemic Model)** and **Model B (The Independent Bernoulli Framework)**.

This analysis aims to demonstrate the "Illusion of Stability". Independent risk frameworks usually appear highly stable because they rely on the assumption that portfolio risks are diversified, however this assumption tends to fall short under broader economic shocks. By evaluating the role of latent asset correlation under a 99% confidence interval, this report demonstrates how ignoring dependency structures severely underestimates tail risk and asset clustering.

2. Data

2.1. Dataset Description

The empirical foundation of this credit risk assessment is built upon two distinctive corporate loan portfolios:

- **Portfolio 1** represents a lower-risk profile. The borrowers in this portfolio have an average baseline Probability of Default of 1,99% and an average Loss Given Default of 47,1%. Risk is evenly spread across moderate outstanding loan sizes.
- **Portfolio 2** represents a higher-risk profile. The average baseline Probability of Default for these borrowers is 3,45% and the average Loss Given Default is 50,3%. This portfolio also features slightly higher average exposure weights (EAD).

2.2. Cleaning Procedures

- **String Standardisation and Structural Correction:** the script standardised inconsistent industry tags into uniform *INDUSTRY_A* or *INDUSTRY_B* tags, and standardised entity codes to a format like: *A-1XPP4*.
- **Row Deduplication and Merging:** To prevent artificial inflation of exposure totals and to eliminate data clutter, the code dropped duplicate entries across portfolios and master files based on the unique *entity_code*, only keeping the first valid instance. For time-series datasets, row reduplication was applied on a joint unique key combining the observation date and the entity/industry label.
- **Missing Fields and Data Imputation:** Blank fields were filled using context-specific metrics to maintain portfolio consistency without skewing trends.
    - **Exposure at Default (EAD):** Incomplete asset balances were filled with each portfolio's median.
    - **Loss Given Default (LGD):** These were filled by grouping borrowers by industry and substituting the sector's median LGD value.

  - **Historical Default Events:** Missing values were assumed to represent non-default years and were set to 0. It was assumed that if a default had happened it would have definitely been kept record.
  - **Baseline Default Probability Hints:** Missing values for default probabilities were handled by grouping historical defaults by industry and mapping the sector's average default frequency.
- **Time-Series Gaps and Pivoting:** The time series returns contained missing value caused by asynchronous trading calendar and holiday closures (assumed). The datasets were sorted chronologically and it was implemented a forward-fill method. This logic carried the assumption that if a trading date was omitted, the stock value remained unchanged. Finally, the industry return dataset was pivoted to transform long-form data rows into distinct industry-wise columns indexed by a calendar date vector.

2.3. Parameter Estimation and Underlying Model Assumptions

The parameter estimation script analyses the historical dataset to map long-run default frequencies across the targeted one-year horizon. The code maps the binary data into an annual timeline. The long-run global default frequency (PD) establishes the structural default barrier. Following Merton's framework, we assume that a firm breaches its obligations if its total value falls below a boundary: $d_i = \Phi^{-1}(\mathrm{PD}_i)$

The systematic asset correlation was calculated directly from the historical default volatility. The script isolated the parameter by calculating it based on: $\rho = \dfrac{\text{annual\_variance}}{\text{global\_pd} \times (1.0 - \text{global\_pd})}$

3. Methodology

3.1. Structural Asset Value Framework (One-Factor) (Model A)

Model A is built around a latent variable framework where a firm's credit state is driven by its underlying asset performance. Since asset value paths are unobservable in real time, the model represents changes in a firm's financial health via a normalised asset return variable, Xi for each borrower i.

To capture the real-world clustering of defaults during macroeconomic downturns, the model breaks asset returns down into a single systematic economic driver and an independent, firm-specific risk component: $Xi = (\sqrt{\rho} \times F) + (\sqrt{1.0 - \rho} \times \epsilon_i)$

In the simulation code, the systematic factor F is drawn as a column vector across 10.000 distinct scenarios, while epsilon is generated as an independent random noise matrix matching the headcount of the portfolio. The total asset return Xi is computed as the weighted combination of these two factors, and a borrower enters breach if Xi <= di.

3.2. Independent Bernoulli Framework (Model B)

Model B serves as a comparative benchmark to illustrate the impact of neglecting systematic risk. This framework completely removes the shared macroeconomic factor (F), operating under the strict assumption that default events are completely independent across firms (rho = 0).

Instead of modelling a continuous asset value path, the simulation triggers default states using a randomised uniform distribution approach. For each of the 10.000 scenarios, the script draws a random probability value from a standard uniform distribution (0,1).

A default is triggered if this independent random draw is less than or equal to the borrower's historical baseline probability of default Ui <= Psi

While both models use the same probability vector (PDi), Model B lacks a shared systemic trigger. As a result, defaults are expected to be scattered across scenarios rather than concentrating into larger scale systemic collapses.

3.3. Risk Metrics

To evaluate tail-risk profiles generated by both models, the simulation routes the resulting loss vectors to calculate three core metrics at a 99% confidence level.

Expected Loss (EL): represents the mathematical average portfolio loss across all simulated paths. It serves as the baseline cost of credit that a financial institution anticipates absorbs through regular earnings.

$$\mathrm{EL} = \frac{1}{N}\sum_{s=1}^{N} \mathrm{Loss}_s$$

Value at Risk (VaR): measures the maximum potential loss that the portfolio will not exceed over a one-year horizon at the specified confidence level (99%). The script extracts this metric by sorting the simulated loss scenarios and identifying the 99th percentile boundary.

$$\mathrm{VaR}_{0.99} = \inf\{L : P(\mathrm{Loss} \le L) \ge 0.99\}$$

Expected Shortfall (ES): assesses the severe tail-risk of the portfolio. It calculates the average loss experienced in the worst 1% of scenarios where the loss has already breached the VaR threshold. This metric highlights the potential severity of insolvency risks during severe market crashes.

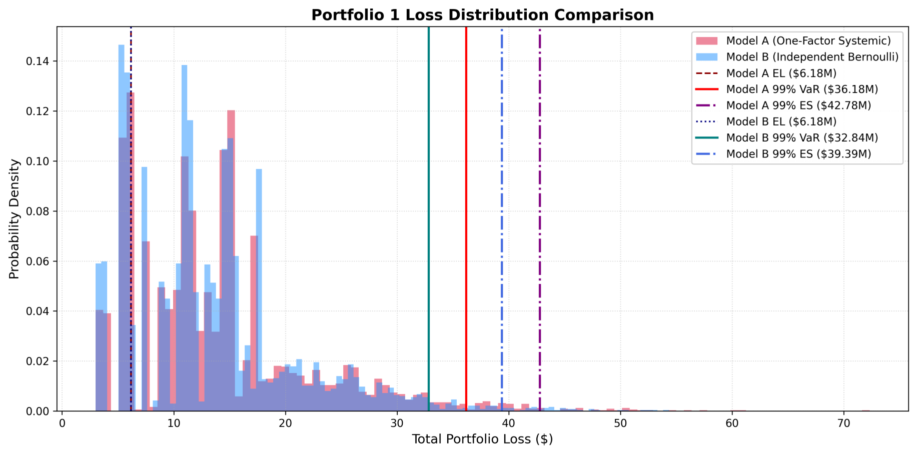
$$\mathrm{ES}_{0.99} = E[\mathrm{Loss} \mid \mathrm{Loss} \ge \mathrm{VaR}_{0.99}]$$

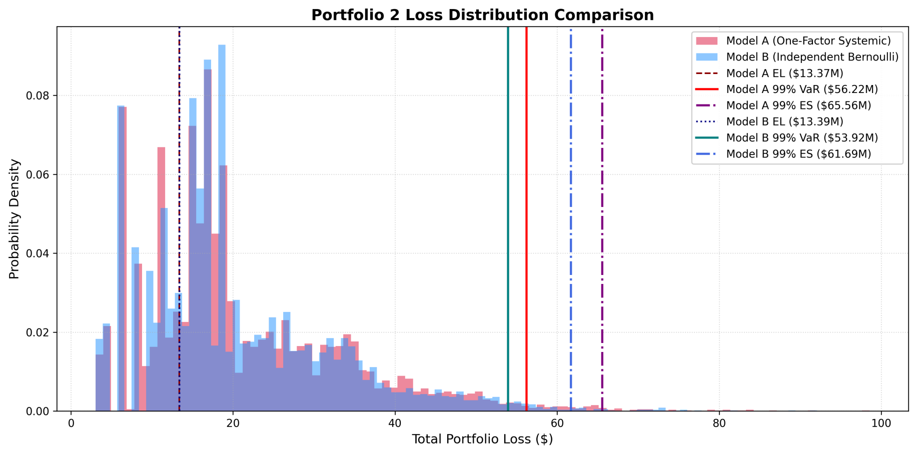
4. Results and Discussion

The corrected results of the risk metrics are shown in the table below, as well as both histograms. To accurately capture the tail shape and prevent scale distortion in the figures, scenarios resulting in zero losses were excluded from the histograms.

Metric	P1 Model A	P1 Model B	P2 Model A	P2 Model B
EL	6.183.582	6.178.187	13.369.694	13.392.199
VaR 99%	36.180.138	32.841.275	56.218.377	53.923.610
ES 99%	42.782.916	39.388.476	65.562.756	61.686.371

Table 1: Risk Measures (in $)





4.1. Core Interpretations

A key mathematical validation of the simulation is that the calculated Expected Loss (EL) values for both portfolios remain almost identical when switching between Model A and Model B. For Portfolio 1, the baseline expected loss centers around **$6.1M** under both setups, while Portfolio 2 stabilises around **$13.1M**.

As visually shown by the histograms, the center of mass for both distributions aligns at the exact same average loss point on the horizontal axis. This occurs because both models pull from the exact same underlying individual credit parameters (PD, EAD, and LGD). Because the total number of simulated scenarios is high (10,000), the long-run arithmetic averages converge on the same expected baseline cost. This proves that introducing systematic correlation (rho) does not alter the baseline, predictable loss expectations of a loan ledger.

When looking across the datasets, Portfolio 2 exhibits more than double the total expected credit risk of Portfolio 1. This is inline with the baseline probability of default that is significantly higher than that of Portfolio 1, as shown before. Consequently, this asset risk inflation directly shifts the entire loss distribution of Portfolio 2 much further down the right-hand axis.

The defining structural insight of this empirical analysis is the massive divergence in the tail metrics (VaR and ES) between the two models, which is clearly illustrated by the shapes of the histograms.

When correlation is turned off (**Model B**), the simulation acts under the assumption that borrower failures are entirely independent. As a result, the distribution curve is tall, narrow, and drops off sharply. Under this framework, defaults are scattered cleanly across all scenarios, meaning the probability of a massive, joint default cluster is mathematically near zero. This independent diversifying effect artificially dampens tail vulnerability, keeping the 99% VaR capped at **$32.84M** for Portfolio 1 and **$53.92M** for Portfolio 2.

When systematic factor dependency is enabled (**Model A**), the latent credit health vectors are anchored to the shared economic driver (F). On the charts, this completely flattens the peak density and pushes a heavy volume of scenarios deep into the right-hand tail. If a major macroeconomic downturn occurs, it pushes multiple corporate asset values below their structural thresholds simultaneously. This dynamic results in severe default clustering. Consequently, Model A reveals a staggering increase in capital risk metrics.

5. Conclusion

This paper successfully executed an end-to-end quantitative credit risk pipeline, transitioning from raw data cleaning and parameter estimation to a simulation. The empirical results confirm that while independent and structural models align perfectly on long-run baseline loss expectations (EL), their representations of tail risk are fundamentally different.

Incorporating a calibrated asset correlation (rho) into Model A flattens the portfolio loss distribution and stretches out a heavy right-hand tail, capturing the real-world reality of default clustering. Portfolio 2's 99% Expected Shortfall breached **$65.56M** under systematic stress, compared to a mere **$61.69M** under the independence assumption. Ultimately, this study demonstrates that accounting for shared systematic risk is not just a regulatory compliance exercise, but a structural necessity to prevent catastrophic undercapitalisation and safeguard financial stability.